3.3.4 Test For Sand

The following are some of the tests conducted to know the quality of sand.

1) Sieve analysis
2) Bulking of sand
3) Voids Ratio
4) Porosity
5) Bulk density



- The chemical name and formula of sand is silicon dioxide and SiO_2.
- Sand is uncountable. It contains particles from 62.5 micron (0.0625mm) to 2mm in diameter.



ACTIVITY 5

Collect some types of sand available nearby your town and display it in your class room.

TDGRH1

3.3.5 Comparison of River Sand & M Sand

S.No	Parameters	River Sand	M-Sand
1	Process	Naturally available on river banks.	Manufactured in factory
2	Shape	Smoother texture with better shape. Demands less water.	Angular and has rougher texture. Angular aggregates demands more water. Water demands can be compensated with cement content.
3	Moisture content	Moisture is trapped in between the particles which is good for concrete purpose.	Moisture is available only in water washed M sand.
4	Concrete Strength	Lesser concrete strength compared to M – sand.	Higher concrete strength compared to river sand.
5	Silt Content	Minimum permissible silt content is 3%.Anything more than 3% is harmful to the concrete's durability. We can expect 5 – 20% silt content in medium quality river sand.	Zero silt. Hence good for construction works.
6	Over sized materials	1 – 6 % of over sized materials can be expected, like pebble, stones, etc.	Since it is artificially manufactured, there is no over sized materials.

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